Ideation Phase
Brainstorm & Idea Prioritization Template

Date	29 January 2026
Team ID	LTVIP2026TMIDS79524
Project Name	Smart Sorting: Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Food spoilage is a major global concern, leading to significant economic loss, health risks, and food waste. In markets, supermarkets, and agricultural supply chains, determining whether fruits are fresh or rotten is often done manually, which can be inconsistent, time-consuming, and prone to human error. Variations in lighting, texture, and appearance make it difficult to accurately assess fruit quality using traditional inspection methods.

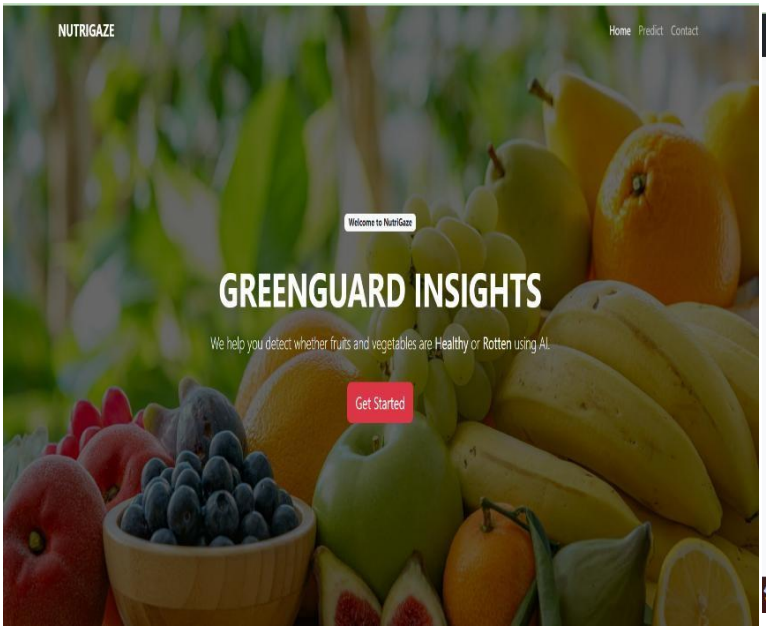
There is a growing need for an intelligent and automated system that can analyze fruit images and reliably distinguish between fresh and spoiled produce. Such a system should be capable of learning visual patterns, reducing dependency on manual inspection, and improving efficiency in quality control processes. It should also provide quick and accurate results through a simple interface accessible to non-technical users.

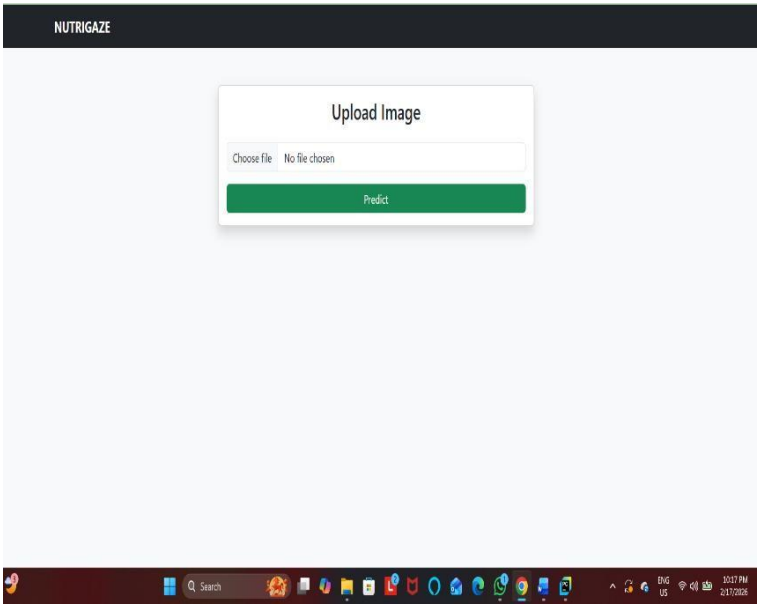
Therefore, the problem is to design and develop a machine learning–based fruit freshness classification system that accurately evaluates fruit condition using image data and delivers instant predictions through a user-friendly web application.

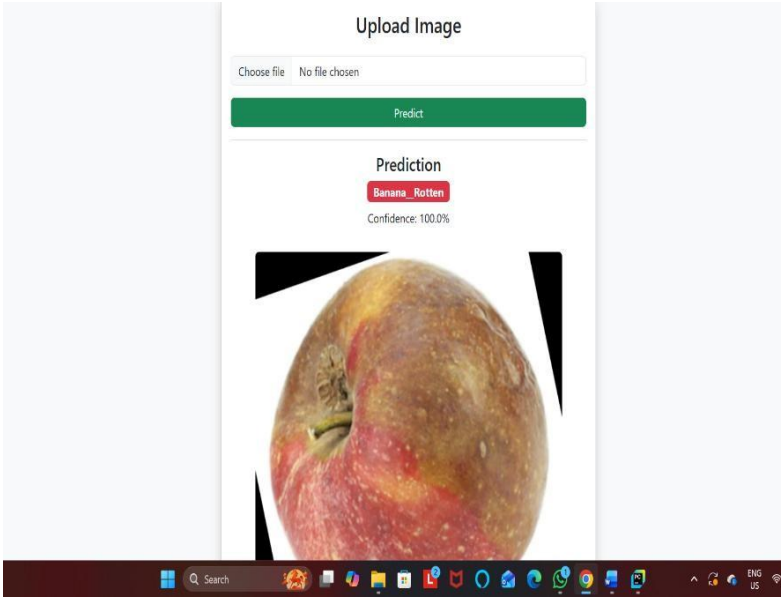
Step-2: Brainstorm, Idea Listing and Grouping

In this stage, the team generated multiple ideas to address the selected fruit freshness classification problem. We discussed different approaches such as developing an automated quality inspection tool, creating a mobile-based fruit scanning application, building a real-time freshness monitoring dashboard, and applying machine learning algorithms for image-based classification. Various image datasets and visual features like color, texture, and surface patterns were considered for model development.

After listing all ideas, we grouped them based on feasibility, impact, and technical implementation. The ideas were categorized into dataset preparation, model training, and application deployment. Finally, we decided to build a machine learning–based fruit freshness classification system integrated into a web application, where users can upload fruit images and receive instant predictions about their freshness.







Step-3: Idea Prioritization

After evaluating different solution approaches, we prioritized developing a deep learning–based web application for fruit freshness classification due to its automation capability, scalability, and practical usability in quality inspection. To implement this solution efficiently, we selected specific technologies and frameworks based on performance, compatibility, and ease of deployment.

For the machine learning model, we chose a convolutional neural network architecture because of its strong performance in image classification tasks and its ability to learn complex visual features. For backend development, we selected Flask (Python) as it provides a lightweight and efficient framework to integrate the trained model with a web interface. For frontend development, we used HTML, CSS, and Bootstrap to create a responsive and user-friendly environment for uploading images and displaying predictions. Additionally, libraries such as TensorFlow/Keras were used for model training, while NumPy and image preprocessing utilities were used to prepare the dataset.

These technologies were prioritized because they are reliable, widely adopted, and well-suited for building an end-to-end deep learning application that delivers accurate and real-time predictions